FOOD SCIENCE AND TECHNOLOGY

Complete 50-Page Course Textbook

Beginner → Intermediate

Food Composition • Processing • Preservation • Safety • Quality • Packaging • Practical Work

A structured course covering the scientific principles of food, food processing technologies, preservation, microbiology, food safety, quality control, packaging, nutrition, sustainability, and product development.

Course Overview

Purpose: This course introduces the scientific and technological principles used to understand, process, preserve, package, evaluate, and safely distribute food.

Learning outcomes: Learners should be able to describe food components, explain spoilage and preservation, identify food hazards, describe major processing operations, explain packaging and quality systems, and outline food-product development.

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Lesson 1: Introduction to Food Science and Technology

Lesson Content

Food science applies biology, chemistry, physics, microbiology, nutrition, and engineering to understand food. Food technology uses this knowledge to process, preserve, package, distribute, and improve food products.

Exercise

Explain the difference between food science and food technology.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 2: Food as a Biological and Chemical System

Lesson Content

Foods contain water, carbohydrates, proteins, fats, vitamins, minerals, pigments, enzymes, and other compounds. Their interactions determine texture, flavor, color, stability, and nutritional quality.

Exercise

Name six major components found in foods.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 3: Water in Foods

Lesson Content

Water strongly affects food texture, chemical reactions, microbial growth, and shelf life. Free water is more available for reactions and microorganisms than water bound to food components.

Exercise

Explain why controlling water is important in food preservation.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 4: Carbohydrates

Lesson Content

Carbohydrates include sugars, starches, and dietary fiber. They provide energy and influence sweetness, viscosity, texture, browning, and gel formation.

Exercise

Differentiate simple sugars, starch, and dietary fiber.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 5: Proteins

Lesson Content

Proteins are polymers of amino acids. In foods they contribute nutrition, structure, emulsification, foaming, gelation, and texture. Heat and processing can change protein structure.

Exercise

Give four technological functions of proteins in foods.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 6: Lipids and Fats

Lesson Content

Lipids contribute energy, flavor, texture, mouthfeel, and help carry fat-soluble vitamins. Oxidation can cause rancidity, so oxygen, light, temperature, and packaging must be controlled.

Exercise

What factors can accelerate lipid oxidation?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 7: Vitamins and Minerals

Lesson Content

Vitamins are organic micronutrients while minerals are inorganic elements needed by the body. Processing can cause losses or changes in their availability.

Exercise

Explain why processing conditions can affect micronutrients.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 8: Enzymes in Food

Lesson Content

Enzymes are biological catalysts. They influence ripening, browning, texture, flavor, and deterioration. Enzyme activity can often be slowed by temperature control, pH adjustment, or other processing methods.

Exercise

Give two useful and two undesirable effects of enzymes in foods.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 9: Food Microbiology

Lesson Content

Microorganisms such as bacteria, yeasts, and molds can be useful in fermentation but can also cause spoilage or foodborne illness. Food microbiology helps control these organisms through hygiene and preservation.

Exercise

Differentiate useful microorganisms from harmful or spoilage microorganisms.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 10: Food Spoilage

Lesson Content

Spoilage can result from microbial growth, enzymatic reactions, chemical changes, physical damage, or environmental conditions. Spoiled food may show changes in odor, appearance, texture, or taste, but absence of obvious spoilage does not guarantee safety.

Exercise

List five causes of food spoilage.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 11: Foodborne Hazards

Lesson Content

Food hazards may be biological, chemical, physical, or allergenic. Prevention relies on hygienic handling, appropriate processing, safe storage, and effective control systems.

Exercise

Give one example of each major hazard category.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 12: Food Hygiene and Sanitation

Lesson Content

Food hygiene covers practices that prevent contamination during production, preparation, processing, storage, and service. Sanitation includes effective cleaning and disinfection procedures.

Exercise

Explain why personal hygiene is important in food handling.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 13: Personal Hygiene for Food Handlers

Lesson Content

Food handlers should maintain clean hands and clothing, avoid contaminating food when ill, protect wounds appropriately, and follow workplace hygiene procedures.

Exercise

List eight good hygiene practices for food handlers.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 14: Cross-Contamination

Lesson Content

Cross-contamination occurs when hazards move from one food, surface, utensil, person, or environment to another. Separation, cleaning, correct storage, and controlled workflows reduce the risk.

Exercise

Give three examples of cross-contamination and how to prevent them.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 15: Food Preservation Principles

Lesson Content

Preservation aims to slow or prevent microbial growth and undesirable chemical or physical changes. Methods include drying, refrigeration, freezing, heating, fermentation, packaging, and approved preservation systems.

Exercise

Why is food preservation important?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 16: Refrigeration

Lesson Content

Refrigeration slows many microbial and chemical processes but does not make food permanently safe. Temperature control, hygienic handling, and suitable storage duration remain important.

Exercise

Explain how refrigeration preserves food.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 17: Freezing

Lesson Content

Freezing converts much of the water in food to ice, reducing available water for microbial growth and slowing reactions. Quality can still change through freezer burn, oxidation, or structural damage.

Exercise

Explain why freezing does not necessarily eliminate microorganisms.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 18: Drying and Dehydration

Lesson Content

Drying reduces water availability and can greatly extend shelf life. Drying methods include sun drying, hot-air drying, spray drying, and other controlled processes.

Exercise

How does dehydration help preserve food?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 19: Heat Processing

Lesson Content

Heat can destroy or reduce microorganisms and inactivate enzymes. Pasteurization and sterilization are examples of thermal processing, but process conditions must be appropriate for the product.

Exercise

Differentiate pasteurization and sterilization conceptually.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 20: Canning

Lesson Content

Canning uses suitable containers and heat processing to produce shelf-stable foods when properly designed and controlled. Safe processing depends on the food, container, acidity, temperature, time, and process design.

Exercise

Why is process control important in canning?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 21: Fermentation

Lesson Content

Fermentation uses microorganisms or their enzymes to transform food components. It can improve flavor, texture, preservation, and sometimes digestibility.

Exercise

Give four examples of foods or beverages produced by fermentation.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 22: Food Additives

Lesson Content

Food additives are substances intentionally used for technological purposes such as preservation, color, texture, acidity control, or stabilization. Their use is subject to safety and regulatory requirements.

Exercise

Name five functional categories of food additives.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 23: Food Color and Pigments

Lesson Content

Food color can come from natural pigments or approved color additives. Processing, pH, oxygen, light, and temperature can affect color stability.

Exercise

Why can food color change during processing?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 24: Flavor and Aroma

Lesson Content

Flavor involves taste, smell, and other sensory sensations. Aroma compounds can be volatile and sensitive to processing and storage conditions.

Exercise

Explain the relationship between taste and aroma.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 25: Sensory Evaluation

Lesson Content

Sensory evaluation uses human senses under controlled conditions to assess attributes such as appearance, aroma, taste, texture, and overall acceptance.

Exercise

What are the main senses used in food sensory evaluation?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 26: Food Texture

Lesson Content

Texture includes properties such as hardness, crispness, chewiness, viscosity, and cohesiveness. It depends on food composition, structure, moisture, and processing.

Exercise

List five texture attributes.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 27: Food Processing Operations

Lesson Content

Processing operations can include sorting, cleaning, size reduction, mixing, heating, cooling, drying, separation, packaging, and storage. Each operation should be selected for its intended effect.

Exercise

Why are unit operations important in food processing?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 28: Cleaning and Sorting

Lesson Content

Raw materials may contain dirt, stones, damaged materials, or other unwanted matter. Cleaning and sorting improve quality and help reduce contamination before later processing.

Exercise

Give four methods used to clean or sort food materials.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 29: Size Reduction

Lesson Content

Cutting, slicing, grinding, milling, and chopping reduce particle size. Size reduction can improve mixing, drying, extraction, cooking, and processing efficiency.

Exercise

State three advantages of size reduction.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 30: Mixing and Emulsification

Lesson Content

Mixing distributes ingredients uniformly. Emulsification helps disperse one immiscible liquid within another, often with emulsifiers or stabilizing systems.

Exercise

Give two examples of food emulsions.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 31: Separation Processes

Lesson Content

Separation may use filtration, centrifugation, pressing, sedimentation, membrane processes, or other techniques to divide food components.

Exercise

Give three examples of food separation processes.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 32: Packaging Technology

Lesson Content

Packaging protects food from contamination, moisture, oxygen, light, physical damage, and loss of quality. It also provides information and supports transport and distribution.

Exercise

List six functions of food packaging.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 33: Food Packaging Materials

Lesson Content

Common packaging materials include glass, metals, paper and paperboard, plastics, and multilayer structures. Selection depends on barrier properties, product characteristics, cost, safety, and sustainability.

Exercise

Compare glass, metal, paperboard, and plastic packaging.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 34: Modified Atmosphere Packaging

Lesson Content

Modified atmosphere packaging changes the gas environment around a product to slow selected deterioration processes. It must be designed for the specific food and storage conditions.

Exercise

Explain the basic principle of modified atmosphere packaging.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 35: Food Quality Control

Lesson Content

Quality control uses measurements and procedures to ensure food meets specified requirements. Tests may cover composition, microbiology, sensory properties, weight, appearance, and packaging integrity.

Exercise

What is the difference between quality control and quality assurance?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 36: Food Safety Management

Lesson Content

Food safety management uses systematic controls to prevent or reduce hazards. HACCP is a preventive approach based on hazard analysis and critical control points.

Exercise

What does HACCP stand for?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 37: HACCP Principles

Lesson Content

HACCP involves hazard analysis, identifying critical control points, establishing critical limits, monitoring, corrective actions, verification, and record keeping.

Exercise

List the seven HACCP principles.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 38: Good Manufacturing Practices

Lesson Content

GMP provides general conditions and practices for hygienic, controlled food production. It can include premises, equipment, personnel, sanitation, documentation, and process controls.

Exercise

List five areas covered by GMP.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 39: Food Laws and Standards

Lesson Content

Food businesses operate under laws, standards, labeling rules, safety requirements, and regulatory oversight. Requirements vary by jurisdiction and product category.

Exercise

Why are food standards important?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 40: Food Labeling

Lesson Content

Labels can communicate product identity, ingredients, allergens, net contents, dates, storage information, manufacturer details, and nutrition information where required.

Exercise

List six types of information commonly found on food labels.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 41: Nutrition and Food Processing

Lesson Content

Processing can improve safety, digestibility, convenience, and shelf life, but some methods can affect nutrients. Food technology seeks suitable conditions that balance safety, quality, nutrition, and practicality.

Exercise

Explain one benefit and one potential nutritional concern associated with processing.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 42: Cereals and Cereal Technology

Lesson Content

Cereals such as maize, rice, wheat, millet, sorghum, and oats are important food crops. Processing may include cleaning, milling, cooking, fermentation, drying, and packaging.

Exercise

Name five cereal crops and one product made from each.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 43: Fruits and Vegetables Technology

Lesson Content

Fruits and vegetables are processed into juices, purees, dried products, frozen products, canned products, jams, and other foods. Color, texture, enzymes, and microbial stability are important concerns.

Exercise

List five processing methods used for fruits or vegetables.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 44: Dairy Technology

Lesson Content

Milk contains water, proteins, fats, lactose, minerals, and vitamins. Dairy processing can include heat treatment, fermentation, separation, homogenization, and packaging.

Exercise

Give five examples of dairy products.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 45: Meat, Fish and Egg Technology

Lesson Content

Animal-derived foods are nutrient-dense but highly perishable. Processing and preservation require strong hygiene, temperature control, appropriate handling, and prevention of cross-contamination.

Exercise

Why do animal-derived foods require careful temperature control?

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 46: Food Waste and Sustainability

Lesson Content

Food waste occurs throughout production, processing, distribution, retail, and consumption. Reduction strategies include better planning, processing efficiency, preservation, packaging, redistribution, and responsible use of by-products.

Exercise

List six ways to reduce food waste.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 47: Food Industry and Careers

Lesson Content

Food science and technology supports careers in quality assurance, production, food microbiology, product development, nutrition-related roles, packaging, research, laboratory analysis, and regulatory work.

Exercise

Name eight careers related to food science and technology.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 48: Food Product Development Case Study

Lesson Content

Product development moves from identifying a need through formulation, processing, prototype testing, sensory evaluation, safety assessment, packaging, shelf-life studies, and controlled production.

Exercise

Outline a development plan for a new fruit-based beverage.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Lesson 49: Final Revision Guide

Lesson Content

Review food composition, microbiology, preservation, processing, packaging, sensory science, quality, safety, HACCP, labeling, sustainability, and major food sectors.

Exercise

Summarize the complete food-processing chain from raw material to consumer.

Lesson checkpoint: Write three key ideas, one question, and one practical example from this lesson.

Final Assessment: 60 Questions and Answers

Attempt each question before reading the answer.

1. What is food science?

Answer: The multidisciplinary study of food composition, properties, reactions, safety, nutrition, and behavior.

2. What is food technology?

Answer: Application of scientific and engineering principles to processing, preservation, packaging, and distribution of food.

3. Name five major food components.

Answer: Water, carbohydrates, proteins, lipids, vitamins, and minerals are major components; dietary fiber and other compounds are also important.

4. Why is water important in foods?

Answer: It affects texture, reactions, microbial growth, and shelf life.

5. What are carbohydrates?

Answer: Food compounds including sugars, starches, and dietary fiber.

6. What are proteins?

Answer: Polymers of amino acids with nutritional and technological functions.

7. What are lipids?

Answer: A group of compounds including fats and oils that contribute energy, texture, flavor, and other functions.

8. What are vitamins?

Answer: Organic micronutrients needed for normal physiological functions.

9. What are minerals?

Answer: Inorganic elements required for physiological functions.

10. What is an enzyme?

Answer: A biological catalyst that speeds specific chemical reactions.

11. What is food microbiology?

Answer: The study of microorganisms associated with food.

12. Name three groups of food microorganisms.

Answer: Bacteria, yeasts, and molds.

13. What is food spoilage?

Answer: Undesirable change that reduces food quality or acceptability.

14. Name four food hazard categories.

Answer: Biological, chemical, physical, and allergenic.

15. What is food hygiene?

Answer: Practices that prevent contamination and support safe food handling.

16. What is sanitation?

Answer: Cleaning and disinfection practices used to maintain hygienic conditions.

17. What is cross-contamination?

Answer: Transfer of hazards from one source to another food, surface, person, or environment.

18. Why preserve food?

Answer: To extend shelf life and maintain safety and quality.

19. How does refrigeration help?

Answer: It slows many microbial and chemical processes.

20. Does freezing sterilize food?

Answer: No. Freezing mainly slows microbial growth and reactions; many microorganisms can survive.

21. How does drying preserve food?

Answer: It reduces water availability and limits microbial growth and reactions.

22. What is pasteurization?

Answer: A controlled heat treatment designed to reduce targeted microorganisms while preserving product quality.

23. What is sterilization?

Answer: A more intensive treatment intended to achieve a much greater reduction of viable microorganisms.

24. What is canning?

Answer: Packaging and heat processing food in suitable containers to achieve stable preservation under defined conditions.

25. What is fermentation?

Answer: Transformation of food components by microorganisms or their enzymes.

26. What are food additives?

Answer: Substances intentionally used for technological purposes in food.

27. Name five additive functions.

Answer: Preservation, coloring, emulsification, stabilization, and acidity regulation.

28. What is sensory evaluation?

Answer: Controlled assessment of food using human senses.

29. Name the main senses used in sensory evaluation.

Answer: Sight, smell, taste, touch, and sometimes hearing.

30. What is food texture?

Answer: The physical and structural properties perceived when food is handled or eaten.

31. What is a unit operation?

Answer: An individual processing step designed to achieve a particular physical or chemical change.

32. Why clean raw materials?

Answer: To remove unwanted matter and reduce contamination.

33. What is size reduction?

Answer: Cutting, grinding, milling, chopping, or similar reduction of particle size.

34. What is emulsification?

Answer: Formation or stabilization of a dispersion of immiscible liquids.

35. Give two food emulsions.

Answer: Mayonnaise and milk.

36. Name three separation processes.

Answer: Filtration, centrifugation, and pressing.

37. What is food packaging?

Answer: A system that contains and protects food and supports handling, distribution, information, and preservation.

38. Name five packaging materials.

Answer: Glass, metal, paperboard, plastics, and multilayer structures.

39. What is modified atmosphere packaging?

Answer: Packaging that changes the gas environment surrounding a food.

40. What is quality control?

Answer: Operational testing and activities used to ensure products meet specifications.

41. What is quality assurance?

Answer: A planned system providing confidence that quality requirements will be consistently achieved.

42. What does HACCP mean?

Answer: Hazard Analysis and Critical Control Points.

43. Name the seven HACCP principles.

Answer: Hazard analysis; critical control points; critical limits; monitoring; corrective actions; verification; record keeping.

44. What is GMP?

Answer: Good Manufacturing Practices for controlled and hygienic food production.

45. Why are food standards important?

Answer: They help protect consumers and establish requirements for safety, quality, composition, and labeling.

46. What information can food labels contain?

Answer: Product name, ingredients, allergens, net contents, dates, storage information, producer information, and nutrition information where required.

47. Can processing affect nutrients?

Answer: Yes. Some processes can cause nutrient losses or changes, while others can improve availability or safety.

48. Name five cereal crops.

Answer: Maize, rice, wheat, millet, and sorghum.

49. Name five dairy products.

Answer: Yogurt, cheese, butter, milk powder, and fermented milk products.

50. Why are many animal foods highly perishable?

Answer: They contain nutrients and moisture that support microbial growth and quality deterioration.

51. What is food waste?

Answer: Food intended for human consumption that is lost, discarded, or not used.

52. Name four ways to reduce food waste.

Answer: Better planning, preservation, efficient processing, improved storage, redistribution, and responsible use of by-products.

53. What is shelf life?

Answer: The period during which food remains acceptable and meets defined quality or safety requirements under stated conditions.

54. What is product development?

Answer: The process of creating, testing, optimizing, and preparing a food product for production and consumers.

55. Why is sensory testing useful in product development?

Answer: It helps determine whether consumers perceive desired characteristics and accept the product.

56. What is a critical control point?

Answer: A step where control can be applied to prevent, eliminate, or reduce a food-safety hazard to an acceptable level.

57. What is a critical limit?

Answer: A measurable criterion that must be met at a critical control point.

58. Why keep food-processing records?

Answer: Records support traceability, verification, quality control, and corrective action.

59. What is traceability?

Answer: Ability to track a food or ingredient through specified stages of production, processing, and distribution.

Practical Exercises

Exercise 1 — Food Composition

Select a common food and identify its major components: water, carbohydrate, protein, fat, minerals, vitamins, and other relevant compounds.

Exercise 2 — Food Preservation

Choose three foods and propose suitable preservation methods. Explain the principle, expected quality changes, and storage conditions.

Exercise 3 — Food Safety Hazard Analysis

Choose a simple food process and identify possible biological, chemical, physical, and allergenic hazards. Propose preventive controls.

Exercise 4 — Sensory Evaluation

Design a simple sensory evaluation form for a food product. Include appearance, aroma, taste, texture, and overall acceptability.

Exercise 5 — Product Development

Develop a concept for a new food product. Describe the target consumer, ingredients, processing steps, packaging, quality tests, shelf-life considerations, and safety controls.

Food Science and Technology Glossary

Food science: Scientific study of food composition, properties, reactions, safety, and behavior.

Food technology: Application of science and engineering to food processing and preservation.

Shelf life: Period during which food remains acceptable under specified conditions.

Spoilage: Undesirable change reducing food quality or acceptability.

Microorganism: Microscopic organism such as a bacterium, yeast, or mold.

Pasteurization: Controlled heat treatment designed to reduce targeted microorganisms.

Sterilization: Intensive treatment intended to achieve a very high reduction of viable microorganisms.

Fermentation: Microbial or enzymatic transformation of food components.

HACCP: Hazard Analysis and Critical Control Points.

GMP: Good Manufacturing Practices.

Sanitation: Cleaning and disinfection practices for hygienic conditions.

Cross-contamination: Transfer of hazards between sources.

Emulsion: Dispersion of one immiscible liquid in another.

Packaging: System that contains and protects food.

Sensory evaluation: Controlled assessment using human senses.

Unit operation: Individual processing step producing a defined effect.

Quality control: Activities used to verify products meet specifications.

Quality assurance: System designed to provide confidence in consistent quality.

Critical control point: Step where food-safety control is essential.

Critical limit: Measurable boundary that must be met at a critical control point.

Traceability: Ability to track food and ingredients through the supply chain.

Food additive: Substance intentionally used for a technological purpose.

Dehydration: Removal of water to improve preservation.

Cold chain: Controlled temperature conditions maintained during storage and distribution.

End of Food Science and Technology

Food science and technology connect scientific knowledge with practical food production. Understanding composition, processing, preservation, safety, quality, packaging, and sustainability provides a strong foundation for further study and professional work.